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(54) **SYNTHESIS OF OMECAMTIV MECARBIL**

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(51) **Int. Cl.**

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C07D 241/04	(2006.01)
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(52) **U.S. Cl.**

CPC ***C07D 401/12*** (2013.01); ***C07D 241/04*** (2013.01)

(58) **Field of Classification Search**

CPC C07D 295/205; C07D 213/75

See application file for complete search history.

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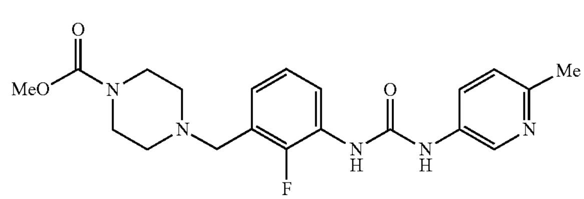
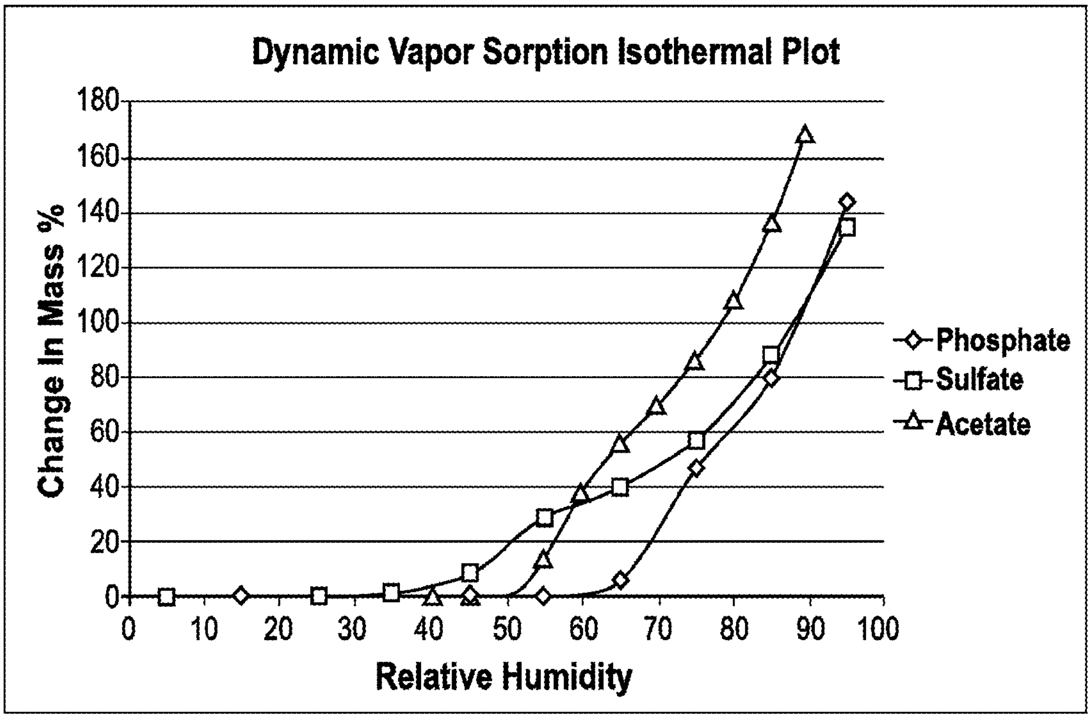
(57) **ABSTRACT**

Provided herein is a synthesis for omecamtiv mecarbil dihydrochloride hydrate and various intermediates. (1)



$*2HCl*H_2O$

13 Claims, 3 Drawing Sheets

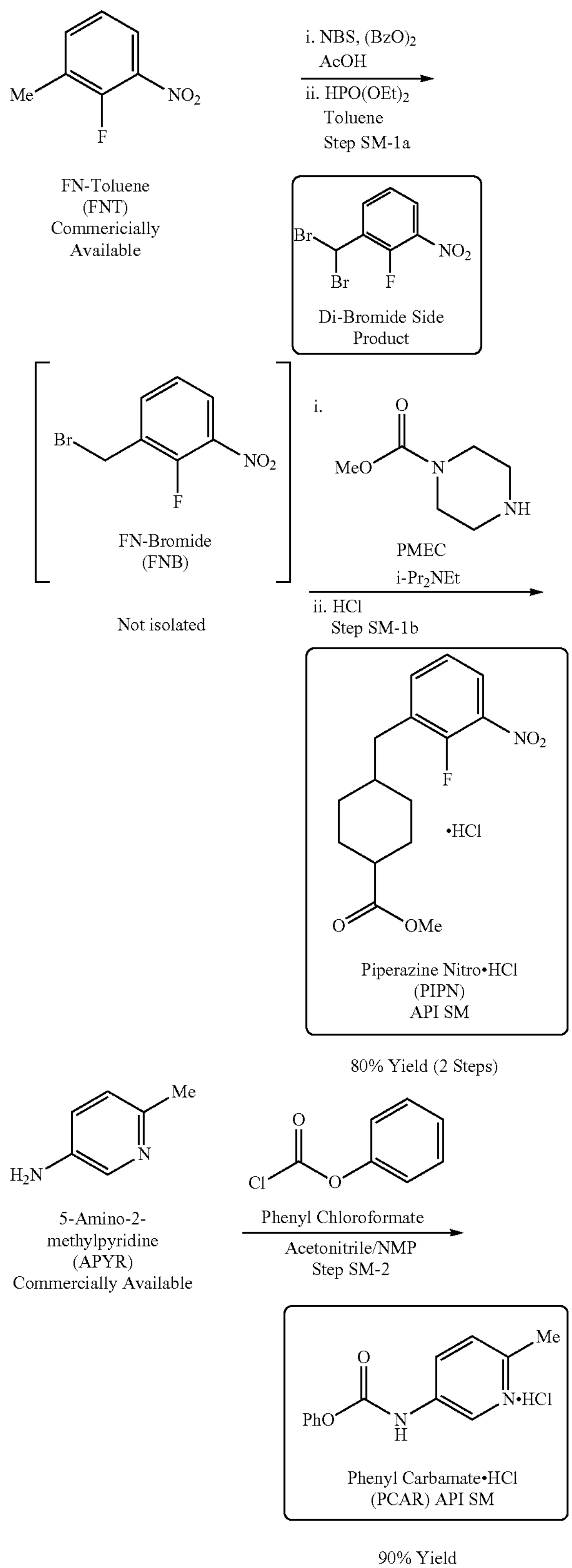


PMEC Salt	DVS Weight Increase Onset
Hemi-Sulfate	35 RH
Acetate	50 RH
Hemi-Phosphate	65 RH

FIGURE 1

chloride hydrate by filtration after wet milling. All the API starting materials are noted in boxes.

Scheme 1



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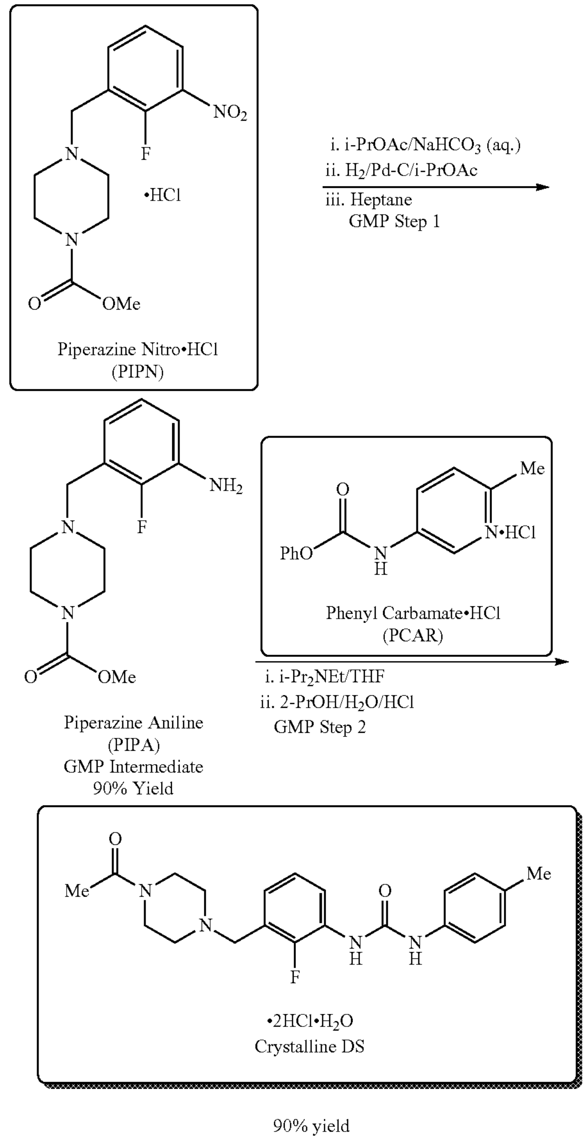
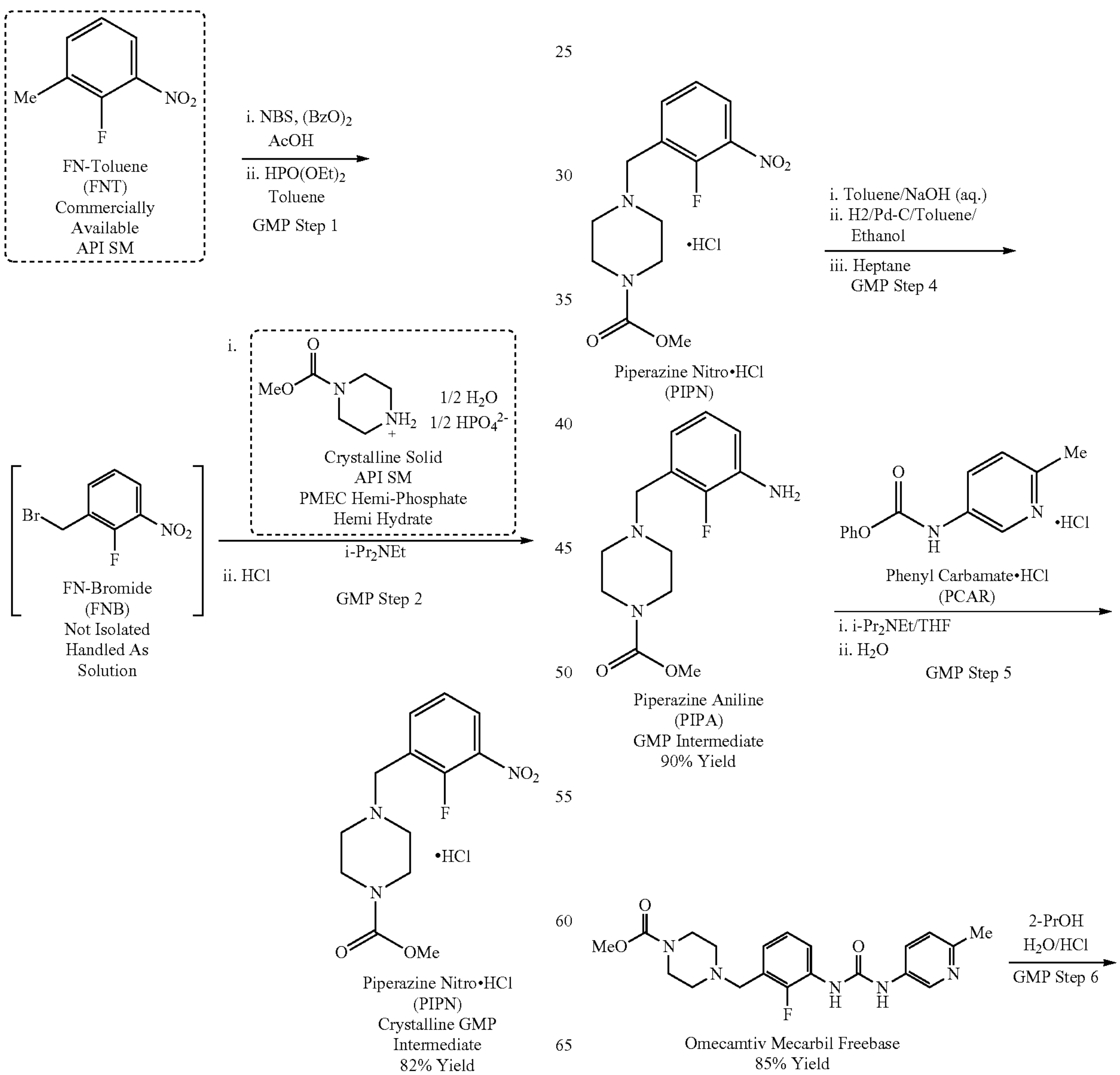


For the synthesis disclosed herein, the API starting materials were moved upstream in the sequence in order to accommodate the requirements for selection and justification of API starting materials to various regulatory bodies, e.g., the EMA and the FDA. As such, the disclosed process herein comprises six steps, compared to the two-step sequence disclosed in WO 2014/152270. This elongated GMP sequence provides several advantages over the shorter sequence. Methyl piperazine-1-carboxylate (PMEC) phosphate is used instead of PMEC free base in the formation of the intermediate piperazine nitro-HCl (PIPN). PMEC free base is an oil that contains various levels of piperazine, which leads to the formation of impurities (e.g., BISN in the product PIPN, see Scheme 3). In contrast, PMEC phosphate is a stable crystalline salt that has low and constant levels of piperazine. Therefore, use of the PMEC hemi-phosphate hemi-hydrate in place of the PMEC free base significantly decreases the formation of impurities. The process disclosed herein also allows for the discontinuation of N-methylpyrrolidinone (NMP) when preparing PCAR, an advantage considering that NMP is difficult to remove and has appeared on REACH protocol lists in the EU (a safety list of chemical materials). In addition, the process disclosed herein alters the solvent in which the hydrogenation of PIPN to generate PIPA is conducted, since the use of isopropyl acetate in the previous process involved an evaporative crystallization operation, which often led to material fouling and inconsistent results. The disclosed process herein replaces a challenging solvent exchange, taking into account the very low solubility of omecamtiv mecarbil freebase in isopropanol (~12 mg/mL) at 20° C. and the formation of an unstirrable slurry during the solvent exchange from tetrahydrofuran (THF) to isopropanol.

The new commercial process disclosed herein to prepare omecamtiv mecarbil dihydrochloride hydrate is shown in Scheme 2. It involves six GMP steps. The designated commercial API starting materials are 2-fluoro-3-nitro-toluene (FNT), 5-Amino-2-methylpyridine (APYR), and PMEC phosphate hydrate.

Scheme 2



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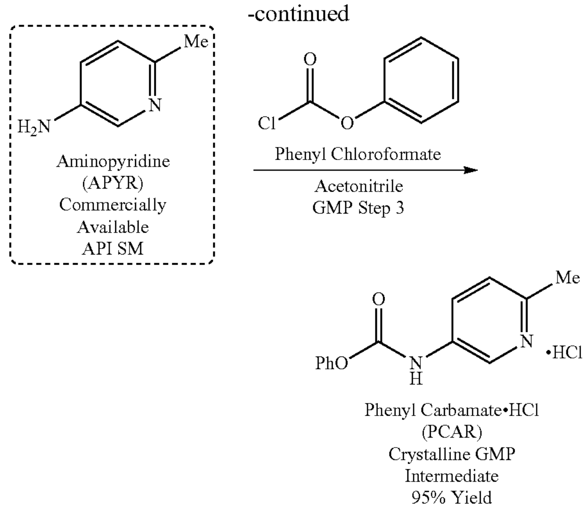

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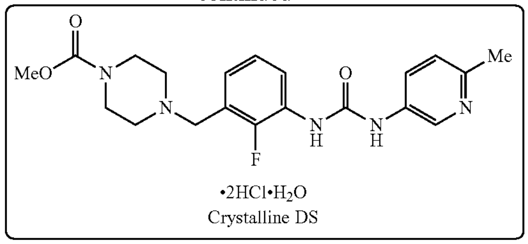
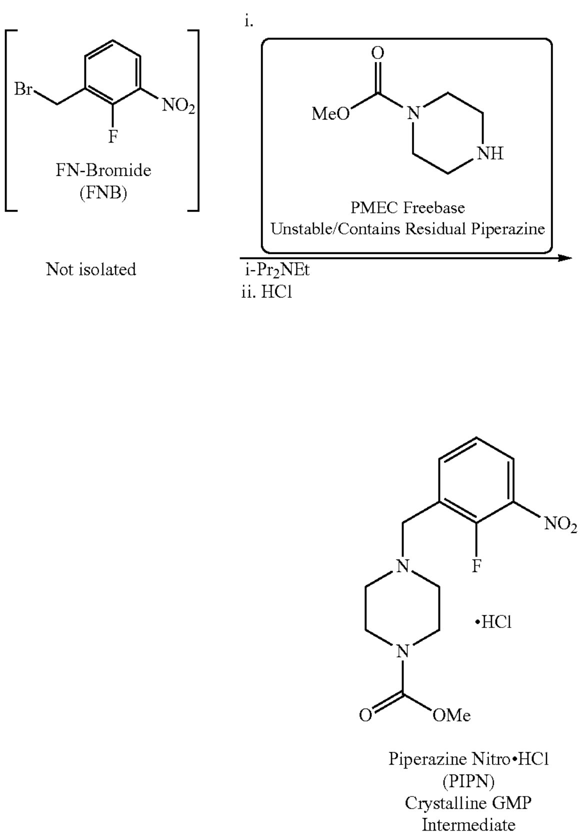


95% Yield

FN-Toluene is a raw material that is manufactured from toluene using a short synthetic sequence. Fractional distillation of the mixture of isomers generated affords the desired regioisomer 2-fluoro-3-nitro-toluene in acceptable purity, with no greater than 0.5 GC area % of any other isomers. 2-Fluoro-3-nitro-toluene (FNT) manufactured using this process has reproducible quality and it can be designated as a commercial API starting material.

PIPN Manufacture: PMEC phosphate, e.g., PMEC phosphate hydrate, is an API starting material prepared in a single step from piperazine. The prior process to prepare PIPN used PMEC freebase as a raw material, which can be purchased, but is an oil that contains various amounts of piperazine. Upon storage at 25° C., piperazine levels up to 18 LC area % were observed in PMEC freebase. As illustrated in Scheme 3, residual piperazine leads to the formation of impurity BISN in product PIPN.

Scheme 3



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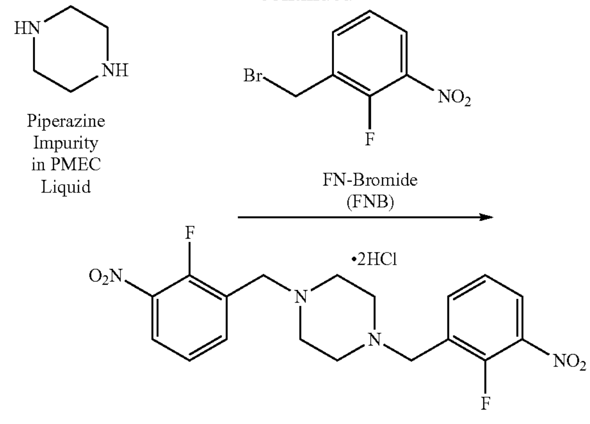

BISN•2HCl Impurity

A stable crystalline salt of PMEC having low and constant levels of piperazine was sought as a commercial API starting material. Multiple salts were thus screened to identify a suitable candidate. PMEC phosphate, e.g., PMEC phosphate hydrate, was found to be less hygroscopic than the corresponding sulfate and acetate salt, as depicted in FIG. **1**. It can be stored in air-sealed aluminum bags to avoid contact with moisture.

As a benefit, PMEC phosphate, e.g., PMEC phosphate hydrate, can be added directly to a reaction mixture to prepare PIPN. By contrast, PMEC acetate has to be converted to PMEC freebase prior to addition to the reaction mixture considering the formation of a side-product from FN-Bromide (FNB) and the acetate anion. PMEC phosphate, e.g., PMEC phosphate hydrate, contains low levels of piperazine (<0.4 GC area %) that do not increase upon storage. PMEC phosphate, e.g., PMEC phosphate hydrate, has successfully been utilized for manufacture of PIPN. The batch of PIPN thus manufactured (5 kg) contained less than 0.1 LC area % of residual BISN.

A process was developed to manufacture PMEC phosphate hydrate involving treatment of piperazine with methyl chloroformate followed by extraction of PMEC as a freebase in the organic layer after neutralization with aqueous sodium hydroxide, as shown in Scheme 4. Subsequent to solvent exchange from dichloromethane to t-butylmethyl ether, the target salt is crystallized by addition of phosphoric acid and filtration. PMEC phosphate hydrate is isolated in 45-50% yield from piperazine and >99 GC area %. Piperazine levels in samples of PMEC phosphate hydrate have been observed to be <0.4 GC area %. DSC spectrum and XRPD pattern for PMEC phosphate hydrate are shown in FIGS. **2** and **3**, respectively. PMEC phosphate has a stoichiometry of about 2:1 of PMEC:phosphate and thus is referenced herein interchangeably as PMEC phosphate, or PMEC hemi-phosphate, PMEC phosphate salt. A hydrate of PMEC phosphate can be formed as detailed herein, and such hydrate has a stoichiometery of about 2:1:1 PMEC:phosphate:water, and is referenced interchangeably as PMEC phosphate hydrate, PMEC hemi-phosphate hemi-hydrate, or PMEC phosphate hydrate. It is understood that the ratio of PMEC, phosphate, and water in the PMEC phosphate hydrate may differ slightly from the 2:1:1 stoichiometric ratio noted above, e.g., to a ratio of 6:4:3 or the like. Elemental analysis and/or single crystal X-ray structural analysis can be performed on the material prepared via the processes disclosed herein. The ratio of the PMEC, phosphate, and water in the isolated salt